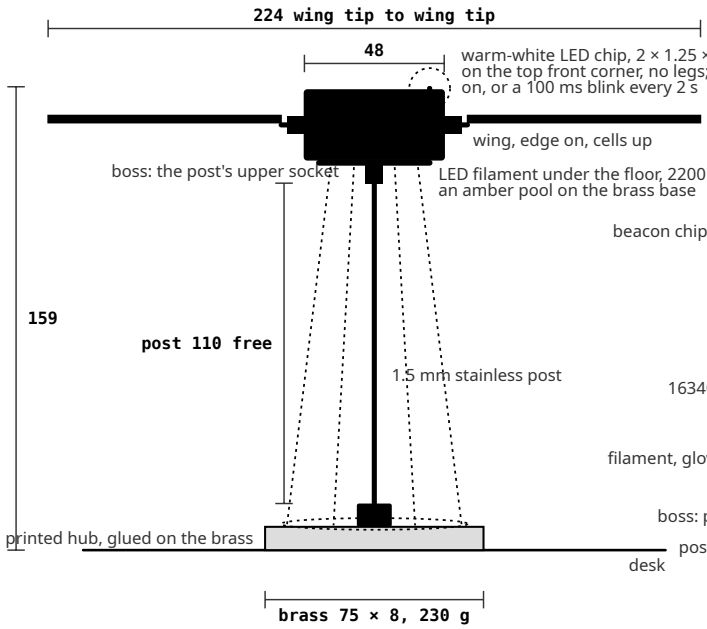
# Desk satellite: how the finished thing looks

A black bus on one stainless post over a brass base, two solar wings, a tiny warm light on the top front corner, an amber pool below. 224 wide × 45 deep × 159 mm tall, about 328 g. Dimensions in mm; each view at its own scale.

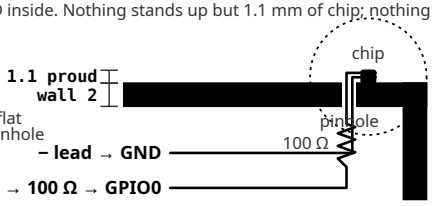
### Front view (lid and USB port on the far side)




### The beacon at the corner, 8 px per mm

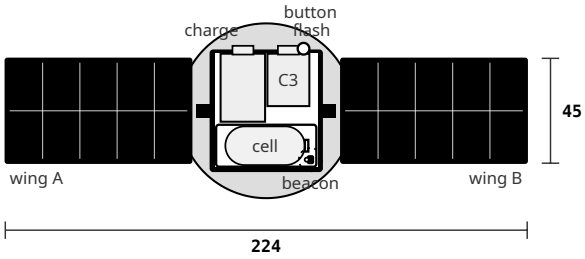
section through the top front-right corner, seen from the front

0805 chip, 2 × 1.25 × 1.1 mm, glued flat on the top 5 mm in from the front and the right face;
its own 0.1 mm enamelled leads run 1 mm over the top and down a 1.2 mm pinhole to the resistor
and GND inside. Nothing stands up but 1.1 mm of chip; nothing is soldered to the box or to steel.



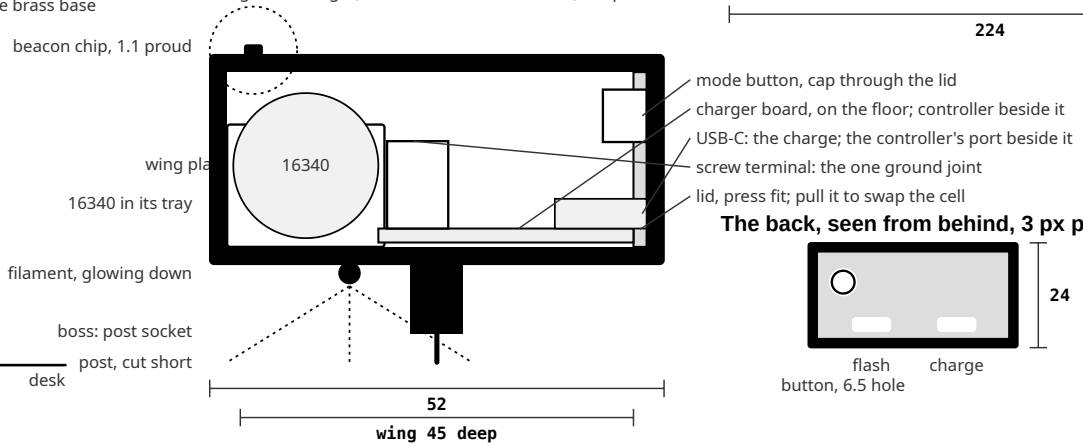

### Top view

front of the desk at the bottom; the brass shows around the bus

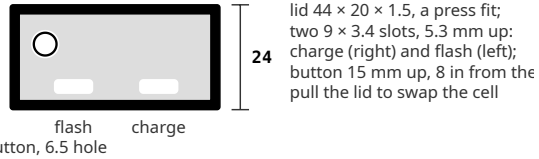



### Side section through the bus, front on the left

cut through the charger; the back is the service side; the post cut short




### The back, seen from behind, 3 px per mm




### Key numbers

| | |
|---|---|
| **224 × 45 × 159 mm** | overall, wings to the top of the LED |
| **48 × 52 × 24 mm** | the bus: printed black, 2 mm walls |
| **2 × 80 × 45 wings, 5 V, series** | 12.2 V in sun, 7 to 8 V in a room |
| **328 g, 230 in the base** | centre of mass 46 mm up; tips at 39° |
| **3 rods, 3 CA sockets** | post and two booms; no frame joints |
| **7 wires, 2 resistors, 1 diode** | 30 AWG; wings and beacon have their own leads |
| **0.22 / 0.42 / 3.4 mA** | asleep / beacon / glow, from the cell |
| **84 / 10 / 2.6 days** | beacon / glow / lamp, no light at all |
| **USB-C 500 mA, 2.2 h** | the fallback charge, through the lid |

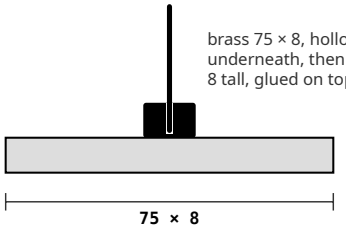
50 mm in the front view

### What the wings deliver (listing figures, scaled by light)

| where | lux | pair, mA | stored a day |
|---|---|---|---|
| desk, room light only | 500 | 0.3 | 0 mAh |
| beside a window, dull day | 1,000 | 0.9 | 1 mAh |
| beside a window, bright day | 3,000 | 2.7 | 12 mAh |
| on the sill, overcast sky | 10,000 | 9.0 | 50 mAh |
| an hour of direct sun | 100,000 | 90.0 | 89 mAh / h |

The charger spends 0.75 mA of input on itself and gives 6 h a day:
'stored' is six hours of what is left. Room light stores nothing.
Beacon mode needs 10 mAh a day, glow 82: a bright window carries the beacon,
the amber pool lives on direct sun or the cable; a USB top-up now and then.

### The base, 3 px per mm




Paper design 2026-11-02, revised twice that morning, approved 2026-11-03 and revised for the 16340 cell. The box is an envelope round the real holder and boards: bench-check before printing. Nothing built.